

Sabancı University

DSA210 Project Final Report

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1. Project Motivation

This project investigates whether directors experience a "fatigue effect" after making long movies. The main research question is that when a director makes a long movie (over 150 minutes), does it negatively affect the rating of his/her next movie?

2. Data Source

2.1 IMDb Dataset

- Source: Kaggle
- Name: IMDb Top 1000 Movies
- Content: 1000 movies, 16 columns
- Used columns: Series_Title, Director, Runtime, IMDB_Rating, Released_Year

2.2 TMDb Dataset

- Source: Kaggle
- Name: TMDb 5000 Movies Dataset
- Content: 5000 movies, 20 columns
- Used columns: original_title, runtime, popularity

2.3 Collection Method

- Both datasets were downloaded as CSV files from Kaggle.
- No API or web scraping was required.
- Data was loaded into Python using pandas and merged on movie titles.

3. Data Analysis

3.1 Data Cleaning and Merging

- Merged IMDb and TMDb on movie title (Series_Title = original_title) using inner join
- Result: 459 common movies
- Converted Released_Year to numeric format
- Removed rows with missing director or year information

3.2 Exploratory Data Analysis (EDA)

- Histograms for runtime, rating, and year distributions
- Correlation matrix between numerical variables
- Identified that most movies are between 90-120 minutes

3.3 Data Preparation for Hypothesis Testing

- Sorted movies by director and year
- Added previous movie information using shift(1) within each director group
- Calculated rating difference: rating_diff = current_rating - previous_rating
- Removed first movies (no previous movie to compare)

3.4 Hypothesis Testing (t-test)

Null Hypothesis (H_0): There is no difference in rating change between directors whose previous movie was long (>150 min) and those whose previous movie was short (≤ 150 min).

Alternative Hypothesis (H_1): Directors who made a long movie experience a greater drop in their next movie's rating.

Groups:

- Long previous movie: $\text{prev_runtime} > 150$ (39 movies)
- Short previous movie: $\text{prev_runtime} \leq 150$ (145 movies)

Method: Independent two-sample t-test

3.5 Machine Learning

Three models were trained and compared using extended features: prev_runtime , prev_rating , year_gap , director_avg_prev , director_exp .

Model	R^2 Score	RMSE
Baseline (Dummy - mean prediction)	-0.002	0.550
Linear Regression	0.464	0.401
Random Forest	0.499	0.388

Random Forest performs slightly better than Linear Regression ($R^2 = 0.499$ vs 0.464). Both models significantly outperform the baseline dummy model ($R^2 = -0.002$), confirming meaningful predictive power. The improved performance of Random Forest suggests the relationship may have some non-linear components that the tree-based model captures better than linear regression.

4. Findings

4.1 Hypothesis Test Results

Group	Number of Movies	Average Rating Change
Long previous movie (>150 min)	39	-0.205
Short previous movie (≤ 150 min)	145	-0.023

p-value = 0.0106 (< 0.05)

The difference is statistically significant.

4.2 Interpretation

The data shows a statistically significant association between longer previous movies and lower subsequent ratings ($p = 0.0106$). Directors who make a movie longer than 150 minutes see their next film's rating drop by 0.205 points on average. Directors with shorter previous movies experience almost no change (-0.023). This is consistent with the director fatigue hypothesis, but does not prove causation.

4.3 Machine Learning Results

Random Forest achieved the highest $R^2 = 0.499$, slightly outperforming Linear Regression ($R^2 = 0.464$). Both models significantly outperform the baseline dummy model ($R^2 = -0.002$), confirming meaningful predictive power.

The improved performance of Random Forest suggests the relationship may have some non-linear components that the tree-based model captures better than linear regression.

Additional features (year_gap, director_avg_prev, director_exp) contributed to the model's predictive ability.

5. Limitations and Future Work

5.1 Limitations

- Sample size: Only 459 movies could be matched between IMDb and TMDb; ML used 89 films after cleaning
- Indirect measurement: "Fatigue" is measured via runtime, not directly observed
- Missing variables: Budget, cast size, genre, and production company were not included
- Top-rated bias: IMDb Top 1000 only includes high-rated movies; results may not generalize to all films
- Small ML dataset: Extended features further reduced available samples

5.2 Future Work

- Include more features: budget, cast size, genre, production company
- Analyze specific director career patterns (e.g., Nolan, Scorsese, Tarantino individually)
- Apply time series analysis to detect recovery periods
- Test with non-English cinema datasets
- Collect larger datasets to better evaluate non-linear models
- Use sentiment analysis on user reviews as an alternative quality metric

6.Conclusion

This project successfully demonstrated that:

- There is a statistically significant association between making long movies and lower subsequent ratings (approximately 0.2 point drop on IMDb's 10-point scale, $p = 0.0106$).
- A Random Forest model using previous movie features (runtime, rating, year gap, director history) can predict rating change with $R^2 = 0.499$, outperforming Linear Regression ($R^2 = 0.464$).
- All models outperform a baseline dummy model, confirming meaningful predictive power.
- The data is consistent with the director fatigue hypothesis. However, this does not prove causation; other factors may also contribute.

7.References

IMDb Top 1000 Movies Dataset – Kaggle

<https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows>

TMDB 5000 Movies Dataset – Kaggle

<https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata>

Python Libraries: pandas, matplotlib, seaborn, scipy, scikit-learn

Project Guidelines: DSA210 Spring 2026